
99% Sure? Gabor Begs to Differ: A Time-Frequency Look at Parkinson’s Handwriting and a Structure-Preserving Image Encoding

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Abstract

Detecting Parkinson’s disease (PD) from handwriting is a clinically useful and difficult problem. PaHaW is a public benchmark of online handwriting from 72 subjects (37 with PD, 35 healthy controls) recorded on a graphics tablet; reported accuracies on it are typically high, often above 90%. We study how much of this performance is supported once the data are evaluated carefully. The PD signature is a time-frequency phenomenon, a 4-6 Hz tremor interleaved with sub-decisecond pen arrests, and the Gabor uncertainty principle ($\sigma_t \sigma_f \geq 1/4\pi$) limits how well any fixed representation can resolve both at once on a short, 150 Hz signal. We (i) review the PaHaW literature and the evaluation choices that can inflate reported scores; (ii) give a time-frequency argument that the task is not yet resolved; and (iii) rather than separating the PD and healthy signals (an ill-posed inverse problem), re-encode the kinematics into a structure-preserving colour image that a frozen Vision Transformer reads. Under repeated nested cross-validation with the Nadeau-Bengio corrected t -test, our best encoding is above chance and above a clinical baseline (F1 0.7457 vs. 0.5080, $p = 0.0187$), but no encoding is statistically separable from its closest rivals. The recovered signal is modest, and the fine-grained ranking is not resolvable at this cohort size.

1 Introduction

Online handwriting is an attractive, non-invasive biomarker for Parkinson’s disease: a tablet records the pen trajectory and pressure while the patient draws, and the trace carries the motor signature of the disease, a 4-6 Hz tremor, reduced and more variable velocity, micrographia, and brief akinetic arrests [5, 7]. A decade of surveys has charted the field [8, 17, 2, 1], and methods range from handcrafted kinematic descriptors to convolutional, recurrent, and reservoir deep models [12, 6, 3]. PaHaW [5] is the standard public benchmark: online handwriting from 72 subjects (37 with PD, 35 controls) on a Wacom tablet. Reported accuracies on it are high, routinely above 90% and sometimes 98-100%. These figures are encouraging; we ask how much survives a careful evaluation.

We find that part of this performance is fragile, for two reasons. The first is *statistical*: on 72 subjects, leakage between train and test, single splits, and naive significance tests across correlated folds all push reported accuracy above what generalizes. The second is *physical*: the PD signature is a time-frequency phenomenon, and the Heisenberg-Gabor uncertainty principle limits how sharply any *fixed* representation can localize the tremor in frequency and the arrests in time at the same instant. A kinematic feature vector, a spectrogram with one window, or a wavelet bank with one quality factor all commit, before seeing the data, to one point on this trade-off; on a few-second recording at 150 Hz that commitment cannot resolve both regimes, so such a score can be optimistic.

Table 1: Selected representative works on PaHaW (best Task 1 result, or the task-merged headline). Family: ♣ handcrafted, ♦ image/CNN, ♥ sequence/RNN. The four audit columns mark nested selection (Nest), freedom from feature-selection bias (FS), subject-disjoint splits (Subj), and reported variability (Var): ✓ yes, ✗ no, ~ unclear. (* task-level LOO; § single hold-out; n/r not reported.)

	Method	Validation	Acc.	F1	Nest	FS	Subj	Var
♣	Drotár et al. [5]	10-fold (tasks 2-8)	0.8130	0.8400~	✗	✗	✓	✗
♣	Impedovo [7]	10-fold	0.9840	n/r	✗	✗	✓	✗
♣	Valla et al. [15] (non-nest.)	10-fold	0.8490	0.8491	✗	✗	✓	✗
♣	Valla et al. [15] (nested)	nested 10-fold	0.7371	0.7300~	✓	✓	✓	✗
♥	Diaz et al. [3]	10-fold	0.9375	n/r	✗	✗	✓	✗
♦	Wang et al. [18]	5-fold (augmented)	0.8470	0.8570	✗	✗	~	✗
♥	Kumar & Ghosh [9]	3:1 hold-out	1.0000§	n/r	✗	✗	~	✗
♣	Shin et al. [13]	LOOCV*	0.9860	0.9860	✗	✗	✗	✗
♦	Ours (effort, ViT, RF)	nested 3×10-fold	0.7524	0.7457	✓	✓	✓	✓

This paper (1) reviews the PaHaW literature and the evaluation choices that can inflate scores (Sec. 2); (2) gives a time-frequency argument that the task is not yet resolved (Sec. 3); and (3) rather than separating the PD and healthy signals (an ill-posed inverse problem), re-encodes the kinematics into a structure-preserving colour image read by a frozen Vision Transformer (Sec. 4). Evaluated carefully (Sec. 5), it clears chance and a clinical baseline but nothing finer.

2 The PaHaW Literature and Its Evaluation Pitfalls

We do not attempt an exhaustive survey. Instead we select a few well-known works that are widely cited as state of the art on PaHaW and that, taken together, represent the main methodological pillars of the field (Table 1). Read side by side, they share four recurring evaluation pitfalls.

Subject leakage. Each PaHaW writer contributes up to eight task recordings. When recordings are pooled, augmented, or held out one at a time, the same patient can appear in train and test, and a classifier may recognize the writer rather than the disease. Shin et al. [13] report 99.98% under leave-one-out, but leave out a single recording while the patient’s other seven stay in training. Group-aware splitting (subject = group) is the standard defense [16], and we enforce it.

Feature-selection bias. Choosing features or hyper-parameters on the full dataset before splitting inflates results. Valla et al. [15] measured this on PaHaW Task 1: non-nested feature selection reports 84.86%, the nested variant 73.71%, an 11.15-point gap from selecting on test data alone.

High performance through feature explosion. A related route to inflated scores is dimensionality: extracting hundreds of descriptors from a few dozen subjects. Shin et al. [13] derive 944 in-air dynamics features for ~75 subjects; with so many degrees of freedom a model can fit the cohort almost perfectly, and combined with non-nested selection this can produce near-ceiling accuracies that need not generalize.

No uncertainty, no valid test. None of the cited works reports a standard deviation or confidence interval; on our 72 subjects the fold-to-fold F1 standard deviation reaches 0.21 (Sec. 5), so a point estimate can coincide with a lucky fold. Tests, when present, are naive paired *t*-tests across folds, which are anti-conservative because folds share training data. Notably, the two studies that nest their selection (Valla’s nested variant and ours) land near 0.74, while the headline figures cluster near 0.99.

3 A Time-Frequency Limit on the Task

The signal. PaHaW records the pen at $f_s = 150$ Hz. The disease superimposes two phenomena at very different scales. The first is the Parkinsonian *tremor*: an involuntary oscillation of the hand, clinically a rest/postural tremor, whose frequency is 4-6 Hz, i.e. the pen position and pressure oscillate

4 to 6 times per second. It is a *narrow-band* event one wants to localize in frequency. The second is a set of aperiodic pen-lifts and akinetic *arrests* of tens of milliseconds, *broadband* impulses one wants to localize in time. A spiral lasts a few seconds, so the tremor contributes only a handful of cycles.

The window dilemma (STFT). The short-time Fourier transform analyses $x[n]$ through a window w of length L ,

$$X[m, k] = \sum_n x[n] w[n - mH] e^{-j2\pi kn/L}, \quad \Delta t \approx \frac{L}{f_s}, \quad \Delta f \approx \frac{f_s}{L}. \quad (1)$$

The window length L sets *both* resolutions at once, with product $\Delta t \Delta f \approx 1$ fixed: a long window resolves frequency but smears the arrests, a short one times the arrests but blurs the tremor. Separating 4 from 6 Hz needs $\Delta f \lesssim 2$ Hz, hence

$$L \gtrsim \frac{f_s}{\Delta f} \approx \frac{150}{2} = 75 \text{ samples} \approx 0.5 \text{ s}, \quad (2)$$

during which any sub-decisecond arrest is washed out. No single L serves both regimes.

The wavelet dilemma. The continuous wavelet transform replaces the fixed window with dilations of a mother wavelet ψ ,

$$W[a, b] = \frac{1}{\sqrt{a}} \sum_n x[n] \psi^*\left(\frac{n-b}{a}\right), \quad Q = \frac{f}{\Delta f} = \text{const.} \quad (3)$$

The window now shrinks with scale, which suits handwriting better, but the quality factor Q and the shape of ψ are fixed before training: the wavelet *moves* the commitment along the bound rather than removing it.

The bound. Both transforms obey the Heisenberg-Gabor uncertainty principle: a signal cannot be arbitrarily sharp in time and frequency at once. For any analysis atom with time spread σ_t and frequency spread σ_f ,

$$\sigma_t \sigma_f \geq \frac{1}{4\pi}, \quad (4)$$

with equality only for the Gabor atom, a Gaussian-modulated sinusoid. Concretely, to pin the tremor to ± 1 Hz one must integrate over $\sigma_t \gtrsim 0.08$ s, longer than an arrest; to time an arrest to ± 10 ms one must accept $\sigma_f \gtrsim 8$ Hz, wider than the tremor band. The two goals are in direct conflict, and (4) fixes the exchange rate.

The limit is upstream of the model. The decisive point is that (4) constrains the *input*, not the classifier. Figure 1 makes it concrete: whatever sits in the black box, a random forest on handcrafted features, an RNN, a CNN, or our own encoder, it receives a short 150 Hz recording whose joint time-frequency content is already limited. The tremor-versus-arrest ambiguity is a property of the measurement, not of the network, so however powerful the model, it cannot recover information the input never contained. A model can be at most as sharp as its input allows; it cannot cheat the physics. The competing feature sets of the literature are competing guesses about where to lose information, none breaks the bound, and a reported 99% therefore need not mean the signal is resolved, only that the evaluation was permissive.

4 Method: a Structure-Preserving Colour Encoding

Dataset and what we train on. PaHaW [5] contains 72 subjects, 37 with PD and 35 healthy controls, recorded on a Wacom Intuos tablet sampling the pen at $f_s = 150$ Hz. Each sample is a 7-tuple

$$s_n = (x_n, y_n, t_n, b_n, p_n, \theta_n^{\text{az}}, \theta_n^{\text{al}}), \quad (5)$$

the pen position (x, y) , timestamp t , contact status $b \in \{0, 1\}$ (on-surface vs. in-air), axial pressure p , and pen azimuth θ^{az} and altitude θ^{al} . From the on-surface samples we derive the kinematics: velocity $v_n = \|(x_{n+1} - x_n, y_{n+1} - y_n)\|/\Delta t$, acceleration $a_n = (v_{n+1} - v_n)/\Delta t$, and *effort* $e_n = v_n p_n$ (a motor-load proxy); together with pressure and the two geometric angles, these are the signals we colour onto the trajectory. Every subject performs eight tasks; we use Task 1, the Archimedean spiral, so each subject contributes one recording and the subject-level decision coincides with the recording-level one, removing the cross-task leakage that inflates other studies.



the input’s time-frequency content is bounded by $\sigma_t \sigma_f \geq 1/4\pi$ *before* the box; no processing inside can resolve the tremor and the arrests beyond what the measurement already allows

Figure 1: The uncertainty bound is upstream of the classifier. Independently of what fills the black box, the pen-to-decision map inherits a time-frequency resolution limit set by the input alone. The difficulty of PaHaW is a property of the measurement, not of the model.

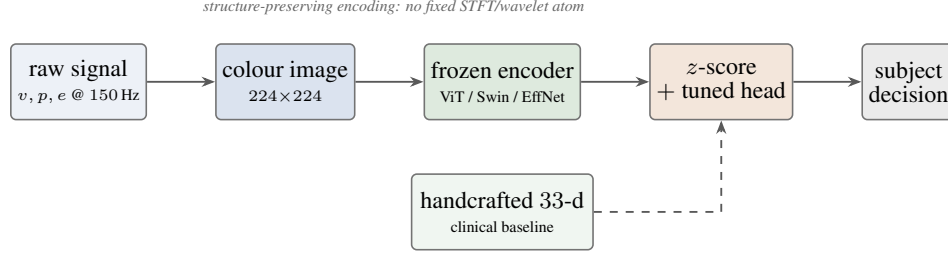


Figure 2: The pipeline. The raw kinematics are re-encoded as a colour image (a structure-preserving quantization), embedded by a frozen backbone, and classified by a per-fold-tuned head. The handcrafted vector (dashed) bypasses the encoder.

Representations. We render ten image modalities of the spiral. Six are single-signal maps that colour the trajectory by one channel with a perceptually uniform colormap (velocity, acceleration, pressure, azimuth, altitude, effort); three are RGB triplets that pack three channels into the colour planes (rgb_vpa, rgb_vpe, rgb_vpz); and one is the raw uncoloured stroke, a geometry-only control. Each image is embedded by one of three frozen ImageNet-pretrained backbones (ViT-B/16 [4], Swin-T [10], EfficientNet-B0 [14]), z -scored, and classified by a per-fold-tuned head (logistic regression, three SVMs, a random forest, two MLPs). The baseline is the 33-d handcrafted clinical vector of Drotár et al. [5] and Impedovo [7] (kinematic and pressure statistics, timing, geometry, and 4-6/0-2 Hz band power). In all, $31 \text{ inputs (10 modalities} \times 3 \text{ backbones} + \text{baseline)} \times 7 \text{ heads} = 217$ cells.

Why re-encode instead of separate. Learning a filter that separates the PD signal from the healthy one is ill-posed at this resolution and cohort size. We change the question: instead of separating, we *preserve*. Drawing the trajectory on a 224×224 canvas and colouring each point by a kinematic channel is a quantization that records *how fast*, *how hard*, and *where* the pen moved, discarding only the fine phase that (4) says we could not have recovered. The encoding keeps the structure that distinguishes a slow, tremulous, high-pressure spiral from a healthy one, and lifts a one-dimensional signal into the two-dimensional input that large-scale visual pretraining is optimized for. This is the dual advantage: it sidesteps the resolution wall and lets us transfer a frozen encoder a 72-subject cohort could never train (Fig. 2).

5 Evaluation

Protocol. We use $R = 3$ repeats (seeds 42, 43, 44) of 10-fold StratifiedGroupKFold, $N = 30$ test folds per cell; an inner 3-fold grid search tunes every head on the training portion only. Models are compared with the Nadeau-Bengio corrected resampled paired t -test [11], which repairs the optimistic variance of the naive fold-wise test by accounting for train/test overlap:

$$t_{\text{NB}} = \frac{\bar{d}}{\sqrt{\left(\frac{1}{N} + \frac{n_{\text{test}}}{n_{\text{train}}}\right) s^2}}, \quad \text{df} = N - 1, \quad (6)$$

Table 2: Champion (= effort, ViT, RF; mean F1 **0.7457**) vs. rivals, naive and Nadeau-Bengio corrected paired t -tests (df= 29). The F1 column is each row model’s *own* mean F1 over the 30 folds (mean AUC in the AUC row), not a difference. red = naive-significant, green = survives correction. Backbone marker: $\diamond$ ViT, $\heartsuit$ Swin. The correction turns the 7 “significant” verdicts here into 0; over all 30 rivals, 29 into 10.

Comparison	F1	t_{naive}	p_{naive}	t_{NB}	p_{NB}
$\diamond$ velocity, ViT, RF	0.7090	1.36	0.1850	0.65	0.5194
$\heartsuit$ raw, Swin, RF	0.6473	2.43	0.0215	1.17	0.2527
$\diamond$ acceleration, ViT, MLP-s	0.6372	3.53	0.0014	1.69	0.1010
$\diamond$ rgb_vpa, ViT, RF	0.6235	2.73	0.0106	1.31	0.1999
$\heartsuit$ rgb_vpz, Swin, MLP-f	0.6230	2.70	0.0114	1.30	0.2046
$\diamond$ rgb_vpz, ViT, RF	0.6226	4.02	0.0004	1.93	0.0631
$\heartsuit$ rgb_vpe, Swin, MLP-f	0.6184	3.18	0.0035	1.53	0.1378
$\heartsuit$ pressure, Swin, MLP-f	0.6157	3.32	0.0025	1.59	0.1221
handcrafted baseline (RF)	0.5080	n/a	<0.0001	2.49	0.0187
champion F1 vs. chance	0.7457	n/a	<0.0001	3.64	0.0011
champion AUC vs. chance	0.7972	n/a	<0.0001	4.11	0.0003

which for 10-fold CV multiplies the naive variance by $0.144/0.033 \approx 4.3$, so t_{NB} is $\sim 2.1 \times$ smaller than its naive twin.

What survives. The best encoding is *effort* on ViT-B/16 with a random forest: $F1 = 0.7457 \pm .18$, $AUC = 0.7972 \pm .19$, $\text{accuracy} = 0.7524 \pm .16$ over 30 folds. It is significantly above chance on every metric (corrected $p \leq 0.0011$) and above the 33-feature clinical baseline (F1 0.7457 vs. 0.5080, corrected $p = 0.0187$; Table 2). This is the part of the signal the theory predicts: *which* channel we encode matters, and effort and velocity, which carry the tremor and its pressure modulation, are the ones that help.

What does not. Nothing finer is resolvable. Against its 30 rivals the winner is significantly better for 29 under a naive test but for only 10 under the correction, and for none of its nine nearest competitors (Table 2). The fold-to-fold spreads overlap almost completely: below the effort/velocity pair the ranking is flat down to the handcrafted baseline, and subject-level F1 is discrete (only 9 distinct values over the ~ 7 held-out subjects of a fold). The best head even flips between random forest (F1) and MLP (AUC = 0.7976) with the metric, and the modal tuned hyper-parameter wins only 48% of folds on average. The picture is two-sided: the encoding recovers a real but coarse signal, effort/velocity > handcrafted > chance, while the fine-grained ranking is not resolvable at this cohort size. This suggests PaHaW is not yet a solved task, and that very high reported figures may partly reflect evaluation choices.

6 Conclusion

The PD signature is a time-frequency object no fixed representation can resolve, by the Heisenberg-Gabor principle, so PaHaW’s high reported figures may partly reflect leakage and optimistic testing rather than a fully solved task. Rather than separate the two signals, we re-encode them into a structure-preserving colour image a frozen vision model can attack; evaluated honestly it clears chance and a clinical baseline but nothing finer. On 72 subjects, the honest number is a modest one.

Reproducibility. Splits use fixed seeds {42, 43, 44}; the sweep, corrected test, and figures are produced by standalone scripts (run_repeated.py, nadeau_bengio.py, full_pairwise.py).

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